

Discrete-Time Linear-Quadratic Dynamic Games

M. Pachter · K.D. Pham

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Abstract Finite-dimensional, time invariant, linear quadratic dynamic games are perhaps the best understood and researched class of dynamic games. This is particularly true for continuous-time linear quadratic differential games. In this paper, the application of the theory of dynamic games to signal processing is considered. We are interested in digital signal processing and therefore we confine our attention to **discrete-time linear-quadratic dynamic games (LQDG)**. In discrete-time the cost function contains product terms between the decision variables which complicates the analysis compared to its continuous-time analogue. With a view to facilitate the application of the theory of dynamic games to digital signal processing, and in particular, disturbance rejection, the complete solution of the discrete-time LQDG is worked out and explicit results are obtained. Thus, discrete-time LQDGs have the distinct advantage of being amenable to analysis, closed-form solutions are possible, and one is in tune with modern digital signal processing techniques. In this paper, minimal necessary and sufficient conditions for the existence of a solution to the discrete-time LQDG are provided and its explicit, closed-form, solution is worked out. This opens the way to designing novel digital signal processing algorithms for disturbance rejection. Information plays a critical role in game theory and in particular in dynamic games. Using our explicit solution of the deterministic LQDG, a hierarchy of three zero-sum stochastic LQDGs characterized by a sequence of information patterns which increase in complexity is analyzed.

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M. Pachter (✉)

Department of Electrical and Computer Engineering, Air Force Institute of Technology AFIT/ENG,
2950 Hobson Way, Bldg. 640, Wright-Patterson Air Force Base, OH 45433, USA
e-mail: Meir.Pachter@afit.edu

K.D. Pham

AFRL/VSSV, 3550 Aberdeen Avenue SE, Kirtland Air Force Base, NM 87117, USA

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1 Introduction

We confine our attention to discrete-time Linear Quadratic Dynamic Games (LQDGs). Discrete-time LQDGs have the distinct advantage of being amenable to analysis, closed-form solutions are possible, and one is in tune with modern digital signal processing techniques. To fix the ideas, the discussion centers on disturbance rejection in control systems and in this tutorial article an active noise cancellation/disturbance rejection scenario [1] using digital signal processing is considered. In Fig. 1 a control system is shown which is excited by a disturbance signal; the control signal is u and the disturbance signal is v . The rejection of disturbances in linear dynamical systems (e.g., linear multi-input multi-output (MIMO) IIR filters/models) is addressed. In conventional filter design, the disturbance is characterized by its frequency content. Thus, frequency domain design methods are routinely used to design low pass filters s.t. the frequency content of the signal of interest is preserved, while the high frequency content, where most of the noise resides, is rejected. In this article, the unwanted disturbance is characterized by its energy. This then mandates that a time domain approach be employed. The control is employed to actively counteract the effects of the disturbance on the system output; therefore, a dynamic game formulation of the problem on hand is called for. The closed-loop control system is shown in Fig. 2. The goal of the disturbance rejection is to be achieved using state feedback. At the same time, the energy of the feedback control signal applied to the filter for the purpose of disturbance rejection should remain within reasonable bounds. Hence, a linear-quadratic dynamic game (LQDG) formulation is employed.

Continuous-time (linear quadratic differential games) are perhaps the best understood dynamic games and have been researched extensively; we refer the reader to [2–8], the monograph [9] and the recent book [10]. In discrete-time, the cost function contains product terms between the decision variables which complicates the analysis compared to the continuous-time analogue. In this paper the general deterministic, finite-dimensional, *discrete-time*, zero-sum, LQDG is carefully analyzed. While the presentation has a tutorial slant, at the same time the complete closed-form solution of the *discrete time* LQDG are provided in conformity with the need to address disturbance rejection/noise cancellation problems using digital signal processing. The

Fig. 1 Dynamical system subject to a disturbance

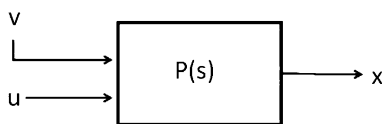
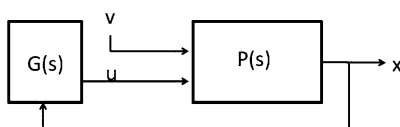


Fig. 2 Disturbance rejection control system



closed-form solution is crucial for the development of digital signal processing algorithms which operate in real time. While steady-state filter performance is most important, the time domain approach allows one to consider also a finite planning horizon and consequently characterize the transient performance of the closed-loop disturbance rejection system.

The disturbance rejection paradigm serves as a vehicle for investigating the impact of information on the solution of dynamic games. Building on our explicit solution of deterministic LQDGs, a hierarchy of three stochastic LQDGs characterized by information patterns which increase in complexity is analyzed.

1.1 Discrete-Time LQDGs

We consider a linear dynamical system,

$$x_{k+1} = Ax_k + Bu_k + Cv_k, \quad x_0 \equiv x_0, \quad k = 0, 1, \dots, N-1, \quad (1)$$

where the state $x_k \in R^n$, the control of the minimizing player $u \in R^{m_u}$ and the control of the maximizing player $v \in R^{m_v}$, A is an $n \times n$ matrix and the minimizing and maximizing players' input matrices B and C are $n \times m_u$ and $n \times m_v$, respectively. The planning horizon is N . An infinite planning horizon is also considered.

The payoff functional is quadratic,

$$J(\{u_k\}_{k=0}^{N-1}, \{v_k\}_{k=0}^{N-1}; x_0) = x_N^T H x_N + \sum_{k=0}^{N-1} (x_{k+1}^T Q x_{k+1} + u_k^T R_u u_k - v_k^T R_v v_k), \quad (2)$$

where H and Q are real symmetric $n \times n$ matrices and the minimizing and maximizing players' control efforts weighting matrices R_u and R_v are typically real symmetric and positive definite $m_u \times m_u$ and $m_v \times m_v$ matrices, respectively. It is oftentimes stipulated that the state penalty matrices H and Q be positive definite or at least positive semidefinite; we shall relax these assumptions as much as possible.

The disturbance signal is v and the controller-generated signal u aims to minimize its effect on the system's output, say $y = \sqrt{Q}x$, while at the same time keeping the control energy expenditure small. The goal of the disturbance v is diametrically opposed. Hence, in our formulation the player/controller, whose control variable is u_k , strives to minimize the payoff functional J , the player/disturbance, whose control variable is v_k , strives to maximize the payoff functional J , and a zero-sum dynamic game is solved.

Concerning the information pattern: Both players are cognizant of the problem parameters, namely, the dynamic matrices (A, B, C) , the payoff functional parameters (H, Q, R_u, R_v) and the planning horizon N ; during the game, at time k , $k = 0, 1, \dots, N-1$, both players have access to the system state x_k .

Alternatively, a dynamic game where the cost functional reflects exclusively the state penalty, that is,

$$J(\{u_k\}_{k=0}^{N-1}, \{v_k\}_{k=0}^{N-1}; x_0) = x_N^T H x_N + \sum_{k=0}^{N-1} x_{k+1}^T Q x_{k+1}$$

and the players available control energy is constrained according to

$$\sum_{k=0}^{N-1} u_k^T R_u u_k \leq 1, \quad \sum_{k=0}^{N-1} v_k^T R_v v_k \leq 1,$$

can be recast into the standard formulation (1) and (2).

In this article, minimal necessary and sufficient conditions for the existence of a solution are derived together with closed-form solutions and explicit formulas for the optimal strategies and the payoff of the *discrete-time* LQDG. The **optimal linear state feedback control laws for both players**, the controller and the deleterious disturbance, are derived explicitly; of course, we are mostly interested in the minimizing player's/controller action, which is conducive to minimizing the cost functional and thus reject the disturbance. Furthermore, minimal conditions for the existence of an optimal solution are provided. This is crucial for the design and implementation of disturbance rejection controllers.

The article is organized as follows. The main result, that is, the explicit closed-form solution of the discrete-time LQDG is given in Sect. 2. The proof of the main result is given in Sect. 3 where a dynamic programming argument is employed to derive the basic recursion and establish necessary and sufficient conditions for the existence of a solution to discrete-time LQDGs. The infinite-horizon discrete-time LQDG is addressed in Sect. 4, where the control law for disturbance rejection in steady-state is synthesized. In Sect. 5 an illustrative example of a disturbance rejection scenario, where the discrete-time LQDG paradigm is invoked, is analyzed. Some fine points germane to dynamic games are discussed. Building on the results of Sects. 2–4, the simplest possible stochastic LQDG is addressed in Sect. 6. In Sect. 7 a novel stochastic LQDG where the players have access to Nature's input and the players strategies are state feedback-stroboscopic strategies, is analyzed. In Sect. 8 a stochastic LQDG with an asymmetric information pattern is analyzed, followed by concluding remarks in Sect. 9.

2 Main Result

A rigorous approach to the disturbance rejection problem entails its formulation as a dynamic game. Thus, the method of dynamic programming (DP) is invoked. It is conducive to the rigorous derivation of necessary and sufficient conditions for the existence of a solution to the dynamic game (1) and (2), and the optimal disturbance rejection strategy can be derived. It turns out that the solution of the *discrete-time* linear quadratic dynamic game is more involved than its continuous time/differential game analogue. At the same time, digital signal processing entails discrete-time and real-time operation; therefore it is imperative to obtain an explicit closed-form solution of the LQDG. The derivation of the said solution has tutorial value and also serves as a good introduction to the theory of dynamic games. The main result, whose proof is given in Sect. 3, is stated in the following theorem.

Theorem 2.1 *A necessary and sufficient condition for the existence of a solution to the discrete-time zero-sum LQDG (1)–(2) is*

$$R_u + B^T P_k B > 0, \quad (3)$$

$$R_v > C^T P_k C, \quad (4)$$

$\forall k = 1, \dots, N - 1$, where the real symmetric matrices P_k are the solution of the difference equation

$$\begin{aligned} P_{k+1} = & A^T \{ P_k - P_k [B S_B^{-1}(P_k) B^T + B S_B^{-1}(P_k) B^T P_k C (R_v - C^T P_k C)^{-1} C^T \\ & + C (R_v - C^T P_k C)^{-1} C^T P_k B S_B^{-1}(P_k) B^T \\ & + C (R_v - C^T P_k C)^{-1} C^T P_k B S_B^{-1}(P_k) B^T P_k C (R_v - C^T P_k C)^{-1} C^T \\ & + C (C^T P_k C - R_v)^{-1} C^T] P_k \} A + Q, \\ P_0 = & H + Q, \quad k = 0, \dots, N - 1. \end{aligned} \quad (5)$$

In (5),

$$S_B(P_k) \equiv B^T P_k B + R_u + B^T P_k C (R_v - C^T P_k C)^{-1} C^T P_k B.$$

In addition, the problem parameters must satisfy the conditions

$$R_u + B^T (Q + H) B > 0, \quad (6)$$

$$R_v > C^T (Q + H) C. \quad (7)$$

The value of the LQDG is

$$V_0(x_0) = x_0^T (P_N - Q) x_0 \quad (8)$$

and the players optimal strategies are the linear state feedback control laws

$$\begin{aligned} u_k^*(x_k) = & -S_B^{-1}(P_{N-k-1}) B^T [I + P_{N-k-1} C (R_v - C^T P_{N-k-1} C)^{-1} C^T] \\ & \times P_{N-k-1} A \cdot x_k, \\ v_k^*(x_k) = & (R_v - C^T P_{N-k-1} C)^{-1} C^T \{ I - P_{N-k-1} B S_B^{-1}(P_{N-k-1}) B^T \\ & \times [I + P_{N-k-1} C (R_v - C^T P_{N-k-1} C)^{-1} C^T] \} P_{N-k-1} A \cdot x_k. \end{aligned} \quad (9)$$

2.1 Discussion

To put things in context, we note that the continuous-time analogue of the difference equation (5) is the much simpler differential Riccati equation (DRE) from the classical theory of continuous-time zero-sum LQDGs,

$$\dot{P} = A^T P + P A - P (B^T R_u^{-1} B - C^T R_v^{-1} C) P + Q, \quad P(0) = H.$$

In conformity with the well-developed theory of continuous-time LQDGs/differential games [2–10], the difference equation (5) is referred to as the difference Riccati equation (DRE). One can appreciate the complexity of the recursion (5), brought about by discrete-time action and yet the solution of the matrix DRE (5) is a crucial step toward obtaining the solution of the discrete-time LQDG. Note however that, for the propagation of P_k , the computation of only an $r_u \times r_u$ and an $r_v \times r_v$ matrix inverse is required and, in the important case of single input systems, this boils down to a scalar division. Furthermore, it is important to recognize that the DRE (5) need not be solved in real-time. The recursion (5) can be solved ahead of time and the matrices P_k can be stored for real-time control action according to the optimal disturbance rejection control law (9). Interestingly, while the continuous-time Riccati equation for control problems has been thoroughly investigated [11], the same cannot be said about the DRE.

Note that, contrary to the continuous-time case, conditions (3) and (4) for the existence of a solution do not automatically mandate that the players real and symmetric control effort weighing matrices R_u and R_v be positive definite. Whereas in continuous-time the players control effort weights must satisfy $R_u \geq 0$, $R_v \geq 0$ (for nonsingular optimal control the conditions are $R_u > 0$, $R_v > 0$), these are not necessary conditions in discrete time. At the same time, and similar to the continuous-time case, conditions (3) and (4) guarantee a solution in pure strategies [12].

We are interested in disturbance rejection and we are therefore interested exclusively in the minimizing player's action u^* , which is given in (9). In the parlance of game theory, it is a pure strategy: a linear state feedback control law is employed. The strategy (9) is a Nash strategy; should the minimizing player unilaterally deviate from this strategy, he will be penalized by incurring a higher cost. Furthermore, the value function $V_0(x_0)$ yields the optimal cost, as a function of the initial state x_0 : since a zero-sum game is solved and a saddle point is obtained, the value function (8) provides a *guarantee* to the minimizing player; that is, as long as the minimizing player sticks with the strategy (9), the actually realized cost will be at most as high as the value $V_0(x_0)$ of the game; most likely, if the maximizing player (a.k.a., the disturbance v) deviates from its optimal play, it might actually be lower. This will indeed be the case: the disturbance is at a disadvantage because, unlike the controller, in reality it does not have access to the system state and

$$J(\{u_k^*(x_k)\}_{k=0}^{N-1}, \{v_k\}_{k=0}^{N-1}; x_0) \leq V_0(x_0) = x_0^T (P_N - Q)x_0, \quad \forall v_k, k = 0, \dots, N-1.$$

3 Proof of Theorem 2.1

The solution of dynamic games entails the synthesis of the players strategies and the calculation of the value function. Since the players have access to complete state information, the players strategies will be state feedback control laws and the solution of the game calls for the application of the method of dynamic programming (DP).

3.1 Dynamic Programming

The value function is denoted by $V_k(x_k)$, where $x_k \in R^n$ and $k = 0, 1, \dots, N - 1$. To solve the LQDG, one sets up the DP recursion

$$\begin{aligned} V_k(x_k) &= \min_{u_k} \max_{v_k} [x_{k+1}^T Q x_{k+1} + u_k^T R_u u_k - v_k^T R_v v_k + V_{k+1}(Ax_k + Bu_k + Cv_k)], \\ V_N(x_N) &= x_N^T H x_N. \end{aligned} \quad (10)$$

Guess. The value function is quadratic in the state,

$$V_k(x_k) = x_k^T P_k x_k, \quad (11)$$

where P_k are real, symmetric, $n \times n$ matrices and $k = 0, 1, \dots, N$.

Inserting the expression (11) into the DP equation (10) yields the boundary condition

$$P_N = H \quad (12)$$

and

$$\begin{aligned} x_k^T P_k x_k &= \min_{u_k} \max_{v_k} [u_k^T R_u u_k - v_k^T R_v v_k \\ &\quad + (Ax_k + Bu_k + Cv_k)^T (Q + P_{k+1})(Ax_k + Bu_k + Cv_k)] \\ &= \min_{u_k} \max_{v_k} \{u_k^T [R_u + B^T (Q + P_{k+1})B] u_k - v_k^T [R_v - C^T (Q + P_{k+1})C] v_k \\ &\quad + 2x_k^T A^T (Q + P_{k+1})Bu_k + 2x_k^T A^T (Q + P_{k+1})Cv_k \\ &\quad + 2u_k^T B^T (Q + P_{k+1})Cv_k\} + x_k^T A^T (Q + P_{k+1})Ax_k, \\ &\forall k = 0, \dots, N - 1. \end{aligned} \quad (13)$$

From the examination of the RHS of (13), we conclude that, for a solution in pure strategies to exist, the following conditions must be invertible: hold:

$$R_u + B^T (Q + P_{k+1})B > 0, \quad (14)$$

$$R_v > C^T (Q + P_{k+1})C, \quad (15)$$

and the following $(m_u + m_v) \times (m_u + m_v)$ blocked matrix

$$M_{k+1} \equiv \begin{bmatrix} B^T (Q + P_{k+1})B + R_u & B^T (Q + P_{k+1})C \\ C^T (Q + P_{k+1})B & C^T (Q + P_{k+1})C - R_v \end{bmatrix}.$$

The matrix M_{k+1} is invertible, provided that the Schur complement [13],

$$\begin{aligned} S_C &\equiv C^T (Q + P_{k+1})C - R_v - C^T (Q + P_{k+1})B [B^T (Q + P_{k+1})B + R_u]^{-1} \\ &\quad \times B^T (Q + P_{k+1})C, \end{aligned}$$

is invertible. Obviously, the Schur complement S_C is a real, symmetric $r_v \times r_v$ matrix and conditions (14) and (15) guarantees that the Schur complement S_C is negative definite. Hence, the Schur complement is invertible and therefore the matrix M_{k+1} is invertible. Thus, we obtain the following proposition.

Proposition 3.1 *The following two conditions are necessary for the existence of a solution in pure strategies to the dynamic game (1)–(2). Conditions (14) and (15) must hold, $\forall k = 0, 1, \dots, N - 1$.*

One then obtains the optimal controls

$$\begin{pmatrix} u_k^* \\ v_k^* \end{pmatrix} = -M_{k+1}^{-1} \begin{bmatrix} B^T \\ C^T \end{bmatrix} (Q + P_{k+1}) A x_k.$$

Inserting this expression into the DP recursion (13), one obtains

$$\begin{aligned} x_k^T P_k x_k &= x_k^T A^T (Q + P_{k+1}) A x_k + x_k^T A^T (Q + P_{k+1}) \begin{bmatrix} B & C \end{bmatrix} M_{k+1}^{-1} \\ &\quad \times \begin{bmatrix} B^T (Q + P_{k+1}) B + R_u & B^T (Q + P_{k+1}) C \\ C^T (Q + P_{k+1}) B & C^T (Q + P_{k+1}) C - R_v \end{bmatrix} M_{k+1}^{-1} \begin{bmatrix} B^T \\ C^T \end{bmatrix} \\ &\quad \times (Q + P_{k+1}) A x_k - 2x_k^T A^T (Q + P_{k+1}) \begin{bmatrix} B & C \end{bmatrix} M_{k+1}^{-1} \begin{bmatrix} B^T \\ C^T \end{bmatrix} \\ &\quad \times (Q + P_{k+1}) A x_k \end{aligned}$$

which yields the backward matrix recursion

$$\begin{aligned} P_k &= A^T [Q + P_{k+1} - (Q + P_{k+1}) \begin{bmatrix} B & C \end{bmatrix} M_{k+1}^{-1} \begin{bmatrix} B^T \\ C^T \end{bmatrix} (Q + P_{k+1})] A, \\ P_N &= H, \quad k = N - 1, \dots, 0. \end{aligned} \quad (16)$$

Upon solving the recursion (16), the game value function

$$V_0(x_0) = x_0^T P_0 x_0$$

is obtained. Furthermore, invoking Proposition 3.1, applying conditions (14) and (15) at $k = N - 1$, and using the boundary condition (12), one concludes that the necessary condition for the existence of a solution in pure strategies is that the problem parameters satisfy

$$R_u + B^T (Q + H) B > 0, \quad (17)$$

$$R_v > C^T (Q + H) C. \quad (18)$$

In the special case of no terminal cost where $H = 0$, the conditions on the problem parameters are

$$R_u + B^T Q B > 0, \quad (19)$$

$$R_v > C^T Q C. \quad (20)$$

In the special case of no state penalty term, where $Q = 0$, the conditions on the problem parameters are

$$R_u + B^T H B > 0, \quad (21)$$

$$R_v > C^T H C. \quad (22)$$

3.2 Explicit Solution of the Discrete-Time LQDG

It is required to calculate the matrix inverse M_{k+1}^{-1} . The Schur complement of the $C^T(Q + P_{k+1})C - R_v$ block in the M_{k+1} matrix is the real, symmetric $r_u \times r_u$ matrix

$$\begin{aligned} S_B(P_{k+1}) &\equiv B^T(Q + P_{k+1})B + R_u \\ &\quad + B^T(Q + P_{k+1})C[R_v - C^T(Q + P_{k+1})C]^{-1}C^T(Q + P_{k+1})B. \end{aligned} \quad (23)$$

Conditions (14) and (15) yield

$$S_B > 0.$$

Thus, the matrix S_B is invertible; therefore, the matrix inverse M_{k+1}^{-1} can be factored as follows:

$$M_{k+1}^{-1} = E D E^T, \quad (24)$$

with

$$\begin{aligned} E^T &= \begin{bmatrix} I & B^T(Q + P_{k+1})C[R_v - C^T(Q + P_{k+1})C]^{-1} \\ 0 & I \end{bmatrix}, \\ D &= \begin{bmatrix} S_B^{-1} & 0 \\ 0 & -[R_v - C^T(Q + P_{k+1})C]^{-1} \end{bmatrix}. \end{aligned}$$

This allows us to obtain explicit formulas for the players linear state feedback control laws,

$$\begin{aligned} u_k^* &= -S_B^{-1}(P_{k+1})B^T\{I + (Q + P_{k+1})C[R_v - C^T(Q + P_{k+1})C]^{-1}C^T\} \\ &\quad \times (Q + P_{k+1})A \cdot x_k, \\ v_k^* &= [R_v - C^T(Q + P_{k+1})C]^{-1}C^T(I - (Q + P_{k+1})BS_B^{-1}(P_{k+1})B^T \\ &\quad \times \{I + (Q + P_{k+1})C[R_v - C^T(Q + P_{k+1})C]^{-1}C^T\})(Q + P_{k+1})A \cdot x_k. \end{aligned}$$

Remark 3.1 Note that the maximizing player control is

$$v_k^* = [R_v - C^T(Q + P_{k+1})C]^{-1}C^T(Q + P_{k+1})(Ax_k + Bu_k^*).$$

Furthermore, using the factorization (24) of the M_{k+1}^{-1} matrix and redefining

$$P_k := P_k + Q, \quad (25)$$

we obtain the explicit backward recursion formula

$$\begin{aligned} P_k = & A^T \{ P_{k+1} - P_{k+1} [BS_B^{-1}(P_{k+1})B^T \\ & + BS_B^{-1}(P_{k+1})B^T P_{k+1} C(R_v - C^T P_{k+1} C)^{-1} C^T \\ & + C(R_v - C^T P_{k+1} C)^{-1} C^T P_{k+1} BS_B^{-1}(P_{k+1})B^T \\ & + C(R_v - C^T P_{k+1} C)^{-1} C^T P_{k+1} BS_B^{-1}(P_{k+1})B^T P_{k+1} C \\ & \times (R_v - C^T P_{k+1} C)^{-1} C^T \\ & + C(C^T P_{k+1} C - R_v)^{-1} C^T] P_{k+1} \} A + Q, \\ P_N = & H + Q, \quad k = N - 1, \dots, 0, \end{aligned} \quad (26)$$

where [see (23) and the definition (25)] the real, symmetric Schur complement $r_u \times r_u$ matrix S_B is the function

$$S_B(P_k) \equiv B^T P_k B + R_u + B^T P_k C(R_v - C^T P_k C)^{-1} C^T P_k B. \quad (27)$$

Finally, it is convenient to reverse the “time arrow” and step forward in time: the forward recursion for P_k [see (5)] is obtained.

4 Disturbance Cancellation in Steady-State

The planning horizon N is chosen by the system designer subject to the constraint

$$N \leq N_{max},$$

where $1 \leq N_{max} \leq \infty$ is exclusively determined by the problem parameters. Indeed, for $1 \leq N_{max}$, the problem parameters must satisfy the minimal conditions (6) and (7). We assume that this is indeed the case so that $1 \leq N_{max}$ and the following discussion is not vacuous.

The maximal length of the planning horizon N_{max} is determined using the theory developed in Sects. 2 and 3.

Algorithm 4.1

Step 0. Let conditions (6) and (7) hold.

Step 1. Iterate the recursion (5), $\forall k = 0, \dots, K - 1$; if $K > 1$, assume that conditions (3) and (4) hold $\forall k = 1, \dots, K - 1$. Set

$$N_{max} = K$$

and calculate P_K .

Step 2. Check conditions (3) and (4) for $k = K$. If conditions (3) and (4) hold, proceed to solve the recursion (5) for P_{K+1} and set

$$N_{max} = K + 1.$$

Set $k = K + 1$ and return to the verification Step 2, where conditions (3) and (4) are checked.

When condition (3) and/or condition (4) fail, the process stops, at which time a finite N_{max} is established.

The DRE (5) is a nonlinear difference equation. When $N_{max} < \infty$, the DRE has a finite escape time and the LQDG (1)–(2) does not have a solution for planning horizons $N > N_{max}$. However, if the process can be continued ad infinitum, e.g., if $\{P_k\}_{k=0}^{\infty}$ is a bounded and monotonic sequence of real, symmetric, positive-definite matrices, then $N_{max} = \infty$. When $N_{max} = \infty$, the LQDG (1)–(2) has a solution $\forall N \geq 1$.

The determination of the longest possible planning horizon N_{max} s.t. a solution of the discrete-time LQDG (1), (2) exists, is an interesting problem in its own right. However, one is especially interested in the infinite-horizon case where $N_{max} = \infty$, namely, the steady-state solution of the DRE. That the solution of the DRE can be extended to $N \rightarrow \infty$ is not at all clear in dynamic games. This comes somewhat as a surprise for, in one-sided optimization problems, that is, in linear-quadratic estimation/Kalman filtering [14], this is achievable under relatively mild assumptions.

Extending the planning horizon of the LQDG to infinite time is most desirable, for then the control action entails linear *constant* gain action; in other words the system is linear time invariant, a most desirable state of affairs in real-time control, digital signal processing, and filtering. Hence, the infinite-horizon discrete-time LQDG warrants special attention. Indeed, steady-state disturbance rejection hinges on the solution [see e.g. (5)] of the following, fairly complex, equation: when $N_{max} = \infty$, $\exists P$ s.t. $\lim_{k \rightarrow \infty} P_k = P$ and the real, symmetric $n \times n$ matrix P is the solution of the algebraic matrix equation

$$\begin{aligned} P = & A^T \{ P - P[BS_B^{-1}(P)B^T + BS_B^{-1}(P)B^T PC(R_v - C^T PC)^{-1}C^T \\ & + C(R_v - C^T PC)^{-1}C^T P BS_B^{-1}(P)B^T \\ & + C(R_v - C^T PC)^{-1}C^T P BS_B^{-1}(P)B^T PC(R_v - C^T PC)^{-1}C^T \\ & + C(C^T PC - R_v)^{-1}C^T]P \} A + Q. \end{aligned} \quad (28)$$

The LQDGs value function is then

$$V_0(x_0) = x_0^T (P - Q)x_0 \quad (29)$$

and the optimal strategies are constant gains, that is, linear time invariant (LTI) control laws. Specifically, the optimal disturbance rejection signal is then generated by the constant gain linear time invariant (LTI) control law

$$u_k^*(x_k) = -S_B^{-1}(P)B^T [I + PC(R_v - C^T PC)^{-1}C^T]PA \cdot x_k, \quad (30)$$

and

$$v_k^*(x_k) = (R_v - C^T P C)^{-1} C^T \{I - P B S_B^{-1}(P) B^T [I + P C (R_v - C^T P C)^{-1} C^T]\} \\ \times P A \cdot x_k.$$

In conformity with the well developed theory of continuous-time linear quadratic differential games [2, 3], the algebraic matrix equation (28) is referred to as the *discrete* algebraic Riccati equation (DARE). Its off-line numerical solution is required for the calculation of the controller's $m_u \times n$ gain matrix and the implementation of steady-state real-time disturbance rejection controls [see e.g. (30)]. The **gain matrix is calculated ahead of time**, implementation is easy, and during real-time operation the execution time is bounded.

A real, symmetric solution P of the DARE (28), if it exists, is not necessarily unique. The correct solution, namely, the solution which is applicable to the LQDG (1)–(2), is the specific solution P of the DARE s.t. the solution P_k of the difference equation (5), initialized according to

$$P_0 = H + Q,$$

satisfies $\lim_{k \rightarrow \infty} P_k = P$. Hence, the following proposition holds.

Proposition 4.1 *A candidate solution P of the DARE (28) applicable to the LQDG (1)–(2), must satisfy the conditions*

$$R_u + B^T P B > 0, \quad R_v > C^T P C.$$

In dynamic games it is not always possible to extend the solution of the DRE, and consequently the solution of the LQDG, to $N \rightarrow \infty$. $N_{\max}$ is obtained by exercising Algorithm 4.1. Even so, and irrespective of whether $N_{\max}$ is finite or infinite, an interesting corollary of Theorem 2.1 is as follows.

Corollary 4.1 *When the initial state is $x_0 = 0$, the value of the game, that is, the minimizing player cost guarantee, is always 0 $\forall N$, provided that the planning horizon $N \leq N_{\max}$.*

Proof This follows from (8) or (11). □

In other words: consider an initially quiescent dynamical system, excited by a persistent disturbance with bounded energy. The disturbance rejection algorithm in Theorem 2.1 provides a zero cost guarantee, irrespective of the length of the planning horizon N -provided that $N \leq N_{\max}$. Thus, the guaranteed cumulative effect of the disturbance on the output is equal to the energy of the disturbance minus the expanded control energy, that is, the applied control energy directly diminishes the effect of the disturbance.

5 Example

An input disturbance rejection scenario is considered. The concepts and algorithms developed in Sects. 2–4 are illustrated using the vehicle of a scalar discrete-time LQDG.

Without loss of generality, let $b = 1$ and $c = 1$. Let $q = 0$, that is, one is exclusively interested in suppressing the effect of the disturbance at the terminal time. The problem parameters are the time constant a , the players control effort weights r_u and r_v , and the terminal state penalty h .

The first-order dynamics is

$$x_{k+1} = ax_k + u_k + v_k \quad (31)$$

and the payoff functional is

$$J(\{u_k\}_{k=0}^{N-1}, \{v_k\}_{k=0}^{N-1}; x_0) = hx_N^2 + \sum_{k=0}^{N-1} (r_u u_k^2 - r_v v_k^2). \quad (32)$$

We first calculate the Schur complement $S_B(P_k)$. Admittedly, in this example where both players' control variables are scalars, the use of the Schur complement merely allows us to avoid the inversion of a 2×2 matrix:

$$S_B(P_k) = \frac{(r_v - r_u)P_k + r_u r_v}{r_v - P_k} > 0.$$

We insert this expression into the P_k recursion formula (5),

$$P_{k+1} = a^2 P_k \left\{ 1 - P_k \left[S_B^{-1} \left(1 + 2P_k \frac{1}{r_v - P_k} + \frac{P_k^2}{(r_v - P_k)^2} \right) - \frac{1}{r_v - P_k} \right] \right\},$$

$$P_0 = h, \quad k = 0, \dots, N - 1$$

and we obtain the scalar nonlinear DRE

$$P_{k+1} = a^2 \frac{P_k}{1 + (\frac{1}{r_u} - \frac{1}{r_v})P_k}, \quad P_0 = h, \quad k = 0, \dots, N - 1. \quad (33)$$

The value function is

$$V_0(x_0) = P_N x_0^2$$

and (9) yields the optimal disturbance rejection strategy

$$u_k^*(x_k) = -a \frac{P_{N-k-1}}{r_u + (1 - \frac{r_u}{r_v})P_{N-k-1}} x_k, \quad k = 0, \dots, N - 1.$$

Concerning the existence of a solution to the scalar discrete-time LQDG (31)–(32), according to Theorem 2.1, the solution of the DRE (33) must satisfy

$$-r_u < P_k < r_v, \quad \forall k = 1, \dots, N - 1$$

In addition, the problem parameters must satisfy

$$-r_u < h < r_v. \quad (34)$$

From (33), we conclude immediately that choosing the state penalty term $h > 0$ and the control effort penalty terms

$$r_v > r_u > 0$$

guarantees that $P_k > 0$, $\forall k = 0, 1, \dots$.

We assume that the problem parameters satisfy the above conditions. Indeed, the requirement that $r_u < r_v$ means that the minimizer energy cost is lower than the maximizer energy cost; in other words, the minimizing player is more energetic and therefore can overcome the action of the disturbance. Hence, $P_k > 0 > -r_u$ and we only need to worry about $P_k < r_v$, $\forall k = 0, \dots, N - 1$.

If in addition the time constant is such that $-1 \leq a \leq 1$, so that the system, which is excited by the disturbance, is stable then P_k is monotonically decreasing and therefore $h < r_v$ guarantees $P_k < r_v$, $\forall k = 0, \dots, N - 1$. Obviously, $N_{\max} = \infty$. If however $|a| > 1$, that is, the open-loop system is not stable, then a finite maximal planning horizon might exist; in other words, it is possible that $N_{\max} < \infty$, in which case the DRE (33) has a finite escape time.

We now proceed to solving the DRE (33). Note that

$$\frac{1}{P_{k+1}} = \frac{1}{a^2} \frac{1}{P_k} + \frac{1}{a^2} \left(\frac{1}{r_u} - \frac{1}{r_v} \right),$$

provided that $P_k \neq 0$. Setting

$$\Pi_k := \frac{1}{P_k}$$

yields the first-order linear difference equation

$$\Pi_{k+1} = \frac{1}{a^2} \Pi_k + \frac{1}{a^2} \left(\frac{1}{r_u} - \frac{1}{r_v} \right), \quad \Pi_0 = \frac{1}{h}$$

whose solution is

$$\Pi_k = \frac{1}{a^{2k}} \left[\frac{1}{h} + \frac{a^{2k} - 1}{a^2 - 1} \left(\frac{1}{r_u} - \frac{1}{r_v} \right) \right],$$

provided that the time constant $|a| \neq 1$. If $a = \pm 1$, we have an arithmetic progression and

$$\Pi_k = \frac{1}{h} + \left(\frac{1}{r_u} - \frac{1}{r_v} \right) k.$$

Thus, the solution of the DRE (33) is the sequence

$$P_k = \begin{cases} \frac{a^{2k}}{\frac{1}{h} + \frac{a^{2k} - 1}{a^2 - 1} \left(\frac{1}{r_u} - \frac{1}{r_v} \right)}, & \forall |a| \neq 1, \\ \frac{1}{\frac{1}{h} + \left(\frac{1}{r_u} - \frac{1}{r_v} \right) k}, & \forall a = \pm 1, \end{cases} \quad (35)$$

and in particular,

$$P_N = \begin{cases} \frac{a^{2N}}{\frac{1}{h} + \frac{a^{2N}-1}{a^2-1}(\frac{1}{r_u} - \frac{1}{r_v})}, & \forall |a| \neq 1, \\ \frac{1}{\frac{1}{h} + (\frac{1}{r_u} - \frac{1}{r_v})N}, & \forall a = \pm 1, \end{cases} \quad (36)$$

so that the value function is

$$V_0(x_0) = \begin{cases} \frac{a^{2N}}{\frac{1}{h} + \frac{a^{2N}-1}{a^2-1}(\frac{1}{r_u} - \frac{1}{r_v})} x_0^2, & \forall |a| \neq 1, \\ \frac{1}{\frac{1}{h} + (\frac{1}{r_u} - \frac{1}{r_v})N} x_0^2, & \forall a = \pm 1. \end{cases} \quad (37)$$

The value of the game is nonnegative, $\forall a \in \mathbb{R}^1$.

When the time constant $|a| > 1$, we must determine N_{max} . To this end, we invoke condition (4) for the existence of a solution,

$$P_k = \frac{a^{2k}}{\frac{1}{h} + \frac{a^{2k}-1}{a^2-1}(\frac{1}{r_u} - \frac{1}{r_v})} < r_v, \quad \forall k = 1, \dots, N_{max} - 1,$$

that is, the following must hold, $\forall k = 1, \dots, N_{max} - 1$:

$$\left(a^2 - \frac{r_v}{r_u}\right)a^{2k} < (a^2 - 1)\frac{r_v}{h} + 1 - \frac{r_v}{r_u}.$$

Now, $\frac{r_v}{r_u} > 1$ and it is readily verifiable that, if

$$1 < |a| \leq \sqrt{\frac{r_v}{r_u}}, \quad (38)$$

then even though the open-loop system is unstable,

$$N_{max} = \infty.$$

If however

$$|a| > \sqrt{\frac{r_v}{r_u}} > 1, \quad (39)$$

then

$$N_{max} = \left\lceil \frac{1}{2} \frac{\log \frac{(a^2-1)\frac{r_v}{h} + 1 - \frac{r_v}{r_u}}{a^2 - \frac{r_v}{r_u}}}{\log |a|} \right\rceil; \quad (40)$$

the notation $\lceil \cdot \rceil$ in (40) designates the largest integer less than or equal to the expression in the square brackets. In this case, when the problem parameters are s.t. $|a| > \sqrt{\frac{r_v}{r_u}}$, the infinite horizon LQDG (31)–(32) does *not* have a solution; this, notwithstanding the fact that the DARE of the LQDG (31)–(32) has a solution: in fact, the DARE has two solutions, namely:

$$P = 0, \quad (41)$$

and

$$P = (a^2 - 1) \frac{r_u r_v}{r_v - r_u}. \quad (42)$$

In summary, when the LQDG parameters $r_v > r_u > 0$, $h > 0$ and the dynamics parameter $|a| \leq 1$, then $N_{max} = \infty$ and the DARE applicable solution is given by (41); and if the dynamics parameter $1 < |a| \leq \sqrt{\frac{r_v}{r_u}}$, also in this case $N_{max} = \infty$, however the DARE's applicable solution is given by (42). In both cases, $|a| \leq 1$, and $1 < |a| \leq \sqrt{\frac{r_v}{r_u}}$, the infinite horizon LQDG (31)–(32) has a solution, the respective solutions (41) and (42) of the DARE apply, and the DRE solution sequence P_k converges to the applicable DARE solution for all initial conditions $0 < h < r_v$. Indeed, when $N \rightarrow \infty$, $P_N \rightarrow P$, where [see also (41) and (42)]

$$P = \begin{cases} \frac{r_u r_v}{r_v - r_u} (a^2 - 1), & \forall \sqrt{\frac{r_v}{r_u}} \geq |a| > 1, \\ 0, & \forall |a| \leq 1 \end{cases} \quad (43)$$

so that the value function is

$$V_0(x_0) = \begin{cases} \frac{r_u r_v}{r_v - r_u} (a^2 - 1) x_0^2, & \forall \sqrt{\frac{r_v}{r_u}} \geq |a| > 1, \\ 0, & \forall |a| \leq 1. \end{cases}$$

Hence, the disturbance rejection controller's steady state optimal control law is

$$u_k^*(x_k) = \begin{cases} -\frac{r_v}{r_v - r_u} (a - \frac{1}{a}) x_k, & k = 0, 1, \dots, \forall \sqrt{\frac{r_v}{r_u}} \geq |a| > 1, \\ 0, & \forall |a| \leq 1, \end{cases}$$

and the dynamics of the closed-loop system excited by the disturbance is

$$x_{k+1} = \begin{cases} \frac{1}{r_v - r_u} (\frac{1}{a} r_v - a r_u) x_k + v_k, & \forall \sqrt{\frac{r_v}{r_u}} \geq |a| > 1, \\ a x_k + v_k, & \forall |a| \leq 1. \end{cases} \quad (44)$$

If the maximizing player plays optimally, the dynamics is

$$x_{k+1} = \begin{cases} \frac{1}{r_v - r_u} (a r_v - \frac{1}{a} r_u) x_k + u_k, & \forall \sqrt{\frac{r_v}{r_u}} \geq |a| > 1, \\ a x_k + u_k, & \forall |a| \leq 1. \end{cases} \quad (45)$$

If both players play optimally, the closed-loop dynamics is

$$x_{k+1} = \begin{cases} \frac{1}{a} x_k, & \forall \sqrt{\frac{r_v}{r_u}} \geq |a| > 1, \\ a x_k, & \forall |a| \leq 1. \end{cases} \quad (46)$$

It is readily verifiable that always [in (44)] the coefficient

$$\left| \frac{1}{r_v - r_u} \left(\frac{1}{a} r_v - a r_u \right) \right| < 1,$$

which guarantees stability. Thus, the minimizer optimal action brings about stable dynamics. In the same vein, it is readily verifiable that in (45) the coefficient

$$\left| \frac{1}{r_v - r_u} \left(ar_v - \frac{1}{a} r_u \right) \right| > 1,$$

that is, the maximizer optimal action tends to destabilize the system, which serves the disturbance well. However, when both players play optimally, the closed-loop dynamics (46) is stable: the more energetic minimizing player ($r_u < r_v$) is able to overcome the maximizer action and always enforce stability. Thus, during optimal play and when $|a| \leq 1$, the trajectory is

$$x_k = a^k x_0, \quad k = 0, 1, \dots,$$

the control energy expended by the minimizing and maximizing players is

$$E_u \equiv r_u \sum_{k=0}^{\infty} u_k^2 = 0$$

and

$$E_v \equiv r_v \sum_{k=0}^{\infty} v_k^2 = 0,$$

respectively; and since $\lim_{N \rightarrow \infty} x_N = 0$, the value function $V_0(x_0) = 0$, $\forall x_0 \in R^1$. When $\sqrt{\frac{r_v}{r_u}} \geq |a| \geq 1$, the trajectory during optimal play is

$$x_k = a^{-k} x_0, \quad \forall k = 0, 1, \dots$$

and the optimal controls are

$$u_k = -\frac{r_v}{r_v - r_u} \left(a - \frac{1}{a} \right) a^{-k} x_0,$$

$$v_k = \frac{r_u}{r_v - r_u} \left(a - \frac{1}{a} \right) a^{-k} x_0.$$

The calculation of the players' control effort, E_u and E_v , entails the summation of a convergent geometric series. We evaluate

$$E_u = \frac{r_u r_v^2}{(r_v - r_u)^2} (a^2 - 1) x_0^2,$$

$$E_v = \frac{r_u^2 r_v}{(r_v - r_u)^2} (a^2 - 1) x_0^2$$

and since $\lim_{N \rightarrow \infty} x_N = 0$, we directly calculate the value of the game as

$$V_0(x_0) = E_u - E_v$$

$$= \frac{r_u r_v}{r_v - r_u} (a^2 - 1) x_0^2,$$

as expected.

For finite planning horizons N , $P_N > 0$ always; in other words, the value of the game is always positive.

When the LQDG parameters $r_v > r_u > 0$, $h > 0$, however, the time constant $|a| > \sqrt{r_v/r_u}$, then $N_{max} < \infty$ and is given by (40); the DRE (33) exhibits a finite escape time, and the DARE's solutions (41) and (42) do *not* apply to the solution of the LQDG (31)–(32).

In general, when the LQDG formulation is used to model pursuit-evasion scenarios [15], it is assumed that the minimizing player/pursuer control effort penalty matrix

$$0 < R_u < R_v.$$

This is tantamount to saying that the pursuer/minimizer control cost is lower than the evader/maximizer, so that the pursuer is more energetic and he will close in on the evader/reduce the miss distance. In light of this observation, in the disturbance rejection scenario, it is interesting to consider the situation where the opposite is the case, that is, in our scalar example $0 < r_v < r_u$. This, in itself, will not preclude the existence of a solution, but will cause the maximum planning horizon to be bounded, that is, $N_{max} < \infty$ always.

6 Stochastic LQDG1 with Complete State Observation

So far, the deterministic LQDG was considered. Building on the results of Sects. 2–4, the simplest possible stochastic LQDG is now addressed and its explicit solution is derived. The case where the dynamics are perturbed by Gaussian process noise is considered—we refer to the linear quadratic Gaussian dynamic game (LQGDG1).

The LQGDG1 dynamics is

$$x_{k+1} = Ax_k + Bu_k + Cv_k + \Gamma w_k, \quad x_0 \equiv x_0, \quad k = 0, 1, \dots, N-1. \quad (47)$$

The process noise input matrix Γ is a $n \times m_w$ matrix and the process noise is

$$w_k \sim N(0, W),$$

where W is a $m_w \times m_w$ real, symmetric, positive-definite matrix.

The payoff is now a random number and therefore the payoff functional is

$$J(\{u_k\}_{k=0}^{N-1}, \{v_k\}_{k=0}^{N-1}; x_0) = E_w \left(x_N^T H x_N + \sum_{k=0}^{N-1} (x_{k+1}^T Q x_{k+1} + u_k^T R_u u_k - v_k^T R_v v_k) \right). \quad (48)$$

The expectation is taken over the process noise sequence $w \equiv \{w_0, \dots, w_{N-1}\}$.

As before, the players have complete state information. Thus, the simplest stochastic linear quadratic dynamic game, LQGDG1, is solved.

6.1 Dynamic Programming

The value function is denoted by $V_k^{(1)}(x_k)$, where $x_k \in R^n$ and $k = 0, 1, \dots, N - 1$. To solve the LQGDG 1, one sets up the DP recursion

$$\begin{aligned} V_k^{(1)}(x_k) = \min_{u_k} \max_{v_k} E_{w_k} ([x_{k+1}^T Q x_{k+1} + u_k^T R_u u_k - v_k^T R_v v_k \\ + V_{k+1}^{(1)}(Ax_k + Bu_k + Cv_k + \Gamma w_k)]), \\ V_N^{(1)}(x_N) = x_N^T H x_N. \end{aligned} \quad (49)$$

Guess. The LQGDG1 value function is quadratic in the state and has the form

$$V_k^{(1)}(x_k) = x_k^T P_k x_k + p_k^{(1)}, \quad (50)$$

where P_k are real, symmetric $n \times n$ matrices and $p_k^{(1)}$ are scalars; $k = 0, 1, \dots, N$.

The $n \times n$ matrices P_k are the same as in the deterministic LQDG addressed in Sects. 2 and 3, as will become apparent in the sequel.

Inserting the expression (50) into the DP equation (49) yields the boundary conditions (12) and

$$p_N^{(1)} = 0. \quad (51)$$

In addition,

$$\begin{aligned} x_k^T P_k x_k + p_k^{(1)} &= \min_{u_k} \max_{v_k} E_{w_k} ([u_k^T R_u u_k - v_k^T R_v v_k \\ &\quad + (Ax_k + Bu_k + Cv_k + \Gamma w_k)^T (Q + P_{k+1}) \\ &\quad \times (Ax_k + Bu_k + Cv_k + \Gamma w_k) + p_{k+1}^{(1)}]) \\ &= \min_{u_k} \max_{v_k} [u_k^T R_u u_k - v_k^T R_v v_k \\ &\quad + (Ax_k + Bu_k + Cv_k)^T (Q + P_{k+1})(Ax_k + Bu_k + Cv_k)] \\ &\quad + \text{Trace}(\Gamma^T (Q + P_{k+1}) \Gamma W) + p_{k+1}^{(1)}, \quad \forall k = 0, \dots, N - 1. \end{aligned} \quad (52)$$

Similar to the analysis in Sect. 3, we obtain the explicit forward recursion formula (5) for P_k , as expected. In addition, the scalar forward recursion

$$p_{k+1}^{(1)} = p_k^{(1)} + \text{Trace}(\Gamma^T P_k \Gamma W), \quad p_0^{(1)} = 0, \quad k = 0, \dots, N - 1, \quad (53)$$

is obtained. In (5) and (53), $P_k := P_k + Q$.

The following theorem holds.

Theorem 6.1 Consider the stochastic control system (47) and payoff functional (48). Assume that, at decision time k , the players have access to the state x_k , that is, the

players have complete state information. Similar to the deterministic case, in the LQGDG1 the players optimal strategies are given by the respective state feedback control laws (9) and

$$v_k^*(x_k) = (R_v - C^T P_{N-k-1} C)^{-1} C^T \{I - P_{N-k-1} B S_B^{-1} (P_{N-k-1}) B^T \\ \times [I + P_{N-k-1} C (R_v - C^T P_{N-k-1} C)^{-1} C^T]\} P_{N-k-1} A \cdot x_k. \quad (54)$$

Thus, the players strategies in the LQGDG1 with complete state information are given by the solution of the deterministic LQDG provided by Theorem 2.1. However, the value function of the zero-sum stochastic game (47)–(48), namely, the players' expected payoff contains an additional term,

$$V_0^{(1)}(x_0) = x_0^T (P_N - Q) x_0 + p_N^{(1)}.$$

The real symmetric matrices P_k are calculated according to the recursion (5) for the deterministic case. Having obtained the sequence $\{P_k\}_{k=0}^N$ the new scalar sequence $p_k^{(1)}$ is calculated according to the recursion (53) so that the value function can be calculated.

As in the deterministic case, conditions (3)–(4) are necessary and sufficient for the existence of a solution to the LQGDG 1 on the planning horizon N .

Remark 6.1 When the planning horizon $N \rightarrow \infty$, that is, the DRE (5) does not have a finite escape time, the players' optimal strategies reach a steady state. At the same time, the scalar recursion (53) approaches an arithmetic progression. Thus, in the LQGDG1, the value function is monotonic and it does *not* reach a steady state.

7 Stochastic LQDG2 with Complete State Observation

A somewhat nonconventional stochastic dynamic game is considered.

The LQGDG (47)–(48) is revisited. As before, the players have complete state information. In addition, at decision time k , the players also have access to Nature's random input w_k , $k = 0, \dots, N - 1$; we refer here to a stroboscopic information pattern. Stochastic optimal control problems where the realization of the random variable is known at decision time are encountered in sequential inspection problems [16]. Stochastic dynamic games with such an information pattern are referred to as LQGDG2.

7.1 Dynamic Programming

The value function is now denoted by $V_k^{(2)}(x_k)$, where $x_k \in R^n$ and $k = 0, 1, \dots, N - 1$. To solve the novel LQGDG2, one sets up the DP recursion

$$V_k^{(2)}(x_k) = E_{w_k} \left(\min_{u_k} \max_{v_k} [x_{k+1}^T Q x_{k+1} + u_k^T R_u u_k - v_k^T R_v v_k \right. \\ \left. + V_{k+1}^{(2)}(A x_k + B u_k + C v_k + \Gamma w_k)] \right), \\ V_N^{(2)}(x_N) = x_N^T H x_N. \quad (55)$$

Remark 7.1 Note that the expectation is now calculated after the optimization operation, whereas in (49) the order is reversed.

Guess. The value function is quadratic in the state and has the form

$$V_k^{(2)}(x_k) = x_k^T P_k x_k + p_k^{(2)}, \quad (56)$$

where P_k are real, symmetric, $n \times n$ matrices and $p_k^{(2)}$ are scalars; $k = 0, 1, \dots, N$.

The $n \times n$ matrices P_k are the same as in the deterministic LQDG addressed in Sects. 2–3 and the stochastic LQGDG1 addressed in Sect. 6, as will become apparent in the sequel.

Inserting the expression (56) into the DP equation (55) yields the boundary conditions (12) and

$$p_N^{(2)} = 0.$$

In addition,

$$\begin{aligned} x_k^T P_k x_k + p_k^{(2)} &= E_{w_k} \left(\min_{u_k} \max_{v_k} [u_k^T R_u u_k - v_k^T R_v v_k + (Ax_k + Bu_k + Cv_k + \Gamma w_k)^T \right. \\ &\quad \left. \times (Q + P_{k+1})(Ax_k + Bu_k + Cv_k + \Gamma w_k)] \right) + p_{k+1}^{(2)}, \\ \forall k &= 0, \dots, N-1. \end{aligned} \quad (57)$$

The optimization in (57) is similar to the optimization in (13) from Sect. 3 where Theorem 2.1 is proven: set $Ax_k := Ax_k + \Gamma w_k$. Hence, the players optimal strategies are the linear state/stroboscopic feedback control laws

$$\begin{aligned} u_k^*(x_k, w_k) &= -S_B^{-1}(P_{N-k-1})B^T[I + P_{N-k-1}C(R_v - C^T P_{N-k-1}C)^{-1}C^T] \\ &\quad \times P_{N-k-1}(Ax_k + \Gamma w_k), \end{aligned} \quad (58)$$

and

$$\begin{aligned} v_k^*(x_k, w_k) &= (R_v - C^T P_{N-k-1}C)^{-1}C^T\{I - P_{N-k-1}BS_B^{-1}(P_{N-k-1})B^T \\ &\quad \times [I + P_{N-k-1}C(R_v - C^T P_{N-k-1}C)^{-1}C^T]\}P_{N-k-1}(Ax_k + \Gamma w_k). \end{aligned} \quad (59)$$

Furthermore, similar to the analysis in Sect. 3 [see e.g. (16)], we calculate

$$\begin{aligned} x_k^T P_k x_k + p_k^{(2)} &= E_{w_k} ((Ax_k + \Gamma w_k)^T (Q + P_{k+1})(Ax_k + \Gamma w_k) \\ &\quad - (Ax_k + \Gamma w_k)^T (Q + P_{k+1}) \begin{bmatrix} B & C \end{bmatrix} M_{k+1}^{-1} \begin{bmatrix} B^T \\ C^T \end{bmatrix} (Q + P_{k+1})] \\ &\quad \times (Ax_k + \Gamma w_k)) + p_{k+1}^{(2)}, \quad k = N-1, \dots, 0. \end{aligned} \quad (60)$$

Hence,

$$\begin{aligned}
 x_k^T P_k x_k + p_k^{(2)} &= x_k^T A^T \left[(Q + P_{k+1}) - (Q + P_{k+1}) [B \ C] M_{k+1}^{-1} \begin{bmatrix} B^T \\ C^T \end{bmatrix} \right. \\
 &\quad \left. \times (Q + P_{k+1}) \right] A x_k \\
 &+ \text{Trace} \left(\Gamma^T \left[(Q + P_{k+1}) - (Q + P_{k+1}) [B \ C] M_{k+1}^{-1} \begin{bmatrix} B^T \\ C^T \end{bmatrix} \right. \right. \\
 &\quad \left. \left. \times (Q + P_{k+1}) \right] \Gamma W \right) + p_{k+1}^{(2)}, \quad k = N-1, \dots, 0. \quad (61)
 \end{aligned}$$

The backward recursion (16) for P_k obtained in the deterministic case is recovered, as expected. Hence, as in the deterministic case, and similar to Theorem 2.1, the matrices P_k are given by the forward recursion (5). In addition, the scalar forward recursion is

$$\begin{aligned}
 p_{k+1}^{(2)} &= p_k^{(2)} + \text{Trace}(\Gamma^T \{ P_k - P_k [B S_B^{-1}(P_k) B^T \\
 &\quad + B S_B^{-1}(P_k) B^T P_k C (R_v - C^T P_k C)^{-1} C^T \\
 &\quad + C (R_v - C^T P_k C)^{-1} C^T P_k B S_B^{-1}(P_k) B^T \\
 &\quad + C (R_v - C^T P_k C)^{-1} C^T P_k B S_B^{-1}(P_k) B^T P_k C (R_v - C^T P_k C)^{-1} C^T \\
 &\quad + C (C^T P_k C - R_v)^{-1} C^T \} P_k) \Gamma W), \\
 p_0^{(2)} &= 0, \quad k = 0, \dots, N-1; \quad (62)
 \end{aligned}$$

in the recursions (5) and (62) $P_k := P_k + Q$.

The following theorem holds.

Theorem 7.1 Consider the stochastic control system (47) and payoff functional (48). Assume that, at decision time k , the players have access to the state x_k and the random disturbance w_k . In the LQGDG2 the players optimal strategies are given by the respective state feedback/stroboscopic control laws (58) and (59). The value function of the zero sum-game (47)–(48) with the stroboscopic information pattern, namely, the players' expected payoff, is

$$V_0^{(2)}(x_0) = x_0^T (P_N - Q) x_0 + p_N^{(2)}.$$

The real symmetric matrices P_k are calculated according to the recursion (5) for the deterministic LQDG; having obtained the sequence $\{P_k\}_{k=0}^N$ the new scalar sequence $p_k^{(2)}$ is calculated according to the new recursion (62).

As in the deterministic case, conditions (3)–(4) are necessary and sufficient for the existence of a solution to the LQGDG 2 on the planning horizon N .

Concerning the scalar difference equation (62): Comparing the matrix expression whose Trace is calculated in (62) and the R.H.S. of (5) we observe that if the dynamics

matrix A is nonsingular (which often is the case in discrete-time control systems), the difference equation (62) can be written in the compact form

$$\begin{aligned} p_{k+1}^{(2)} &= p_k^{(2)} + \text{Trace}(\Gamma^T (A^{-1})^T (P_{k+1} - Q) A^{-1} \Gamma W)), \\ p_0^{(2)} &= 0, \quad k = 0, \dots, N-1. \end{aligned}$$

8 Stochastic LQDG3 with Complete State Observation

The following stochastic dynamic game with an asymmetric information pattern is now considered.

The LQGDG (47)–(48) is revisited. As before, the players have complete state information. In addition, at decision time k the P player, namely, the minimizing player whose control input is u_k , also has access to Nature's random input w_k , $k = 0, \dots, N-1$. The E player, namely, the maximizing player whose control input is v_k , does not have access to w_k . Thus, E's information pattern is as in Sect. 6 whereas P's information pattern is as in Sect. 7; evidently, player P has an information advantage: player P knows everything that player E knows and more. Although the information pattern in the novel LQGDG 3 is asymmetric, the problem is tractable.

8.1 Dynamic Programming

The value function of P is denoted by $V_k^{(P)}(x_k)$, where $x_k \in R^n$ and $k = 0, 1, \dots, N-1$. To solve the novel LQGDG3, player P sets up his DP recursion

$$\begin{aligned} V_k^{(P)}(x_k) &= E_{w_k} \left(\min_{u_k} [x_{k+1}^T Q x_{k+1} + u_k^T R_u u_k - v_k^T R_v v_k \right. \\ &\quad \left. + V_{k+1}^{(P)}(A x_k + B u_k + C v_k + \Gamma w_k)] \right), \\ V_N^{(P)}(x_N) &= x_N^T H x_N. \end{aligned} \quad (63)$$

The value function of E is denoted by $V_k^{(E)}(x_k)$, where $x_k \in R^n$ and $k = 0, 1, \dots, N-1$. Player E sets up his DP recursion

$$\begin{aligned} V_k^{(E)}(x_k) &= \max_{v_k} \{ E_{w_k} ([x_{k+1}^T Q x_{k+1} + u_k^T R_u u_k - v_k^T R_v v_k \\ &\quad + V_{k+1}^{(E)}(A x_k + B u_k + C v_k + \Gamma w_k)]) \}, \\ V_N^{(E)}(x_N) &= x_N^T H x_N. \end{aligned} \quad (64)$$

Remark 8.1 Note that in (63)–(64), the expectation and optimization operations are reversed.

Guess. The players' value functions are quadratic in the state and are of the form

$$V_k^{(P)}(x_k) = x_k^T P_k^{(P)} x_k + p_k^{(P)}, \quad (65)$$

$$V_k^{(E)}(x_k) = x_k^T P_k^{(E)} x_k + p_k^{(E)}, \quad (66)$$

where $P_k^{(P)}$ and $P_k^{(E)}$ are real, symmetric $n \times n$ matrices, and $p_k^{(P)}, p_k^{(E)}$ are scalars, $k = 0, 1, \dots, N$.

Take P's point of view. Inserting the expression (65) into his DP equation (63), P obtains the boundary conditions

$$\begin{aligned} P_N^{(P)} &= H, \\ p_N^{(P)} &= 0. \end{aligned}$$

In addition,

$$\begin{aligned} x_k^T P_k^{(P)} x_k + p_k^{(P)} &= E_{w_k} \left(\min_{u_k} [u_k^T R_u u_k - v_k^T R_v v_k + (Ax_k + Bu_k + Cv_k + \Gamma w_k)^T \right. \\ &\quad \times (Q + P_{k+1}^{(P)})(Ax_k + Bu_k + Cv_k + \Gamma w_k)] \\ &\quad \left. + p_{k+1}^{(P)}, \quad \forall k = 0, \dots, N-1. \right) \end{aligned} \quad (67)$$

Upon performing the minimization operation and provided that

$$R_u + B^T(Q + P_k^{(P)})B > 0, \quad \forall k = 1, \dots, N, \quad (68)$$

holds, P concludes that his optimal control u_k^* must satisfy the linear equation

$$[R_u + B^T(Q + P_{k+1}^{(P)})B]u_k^* + B^T(Q + P_{k+1}^{(P)})Cv_k = -B^T(Q + P_{k+1}^{(P)})(Ax_k + \Gamma w_k). \quad (69)$$

Next, take E's point of view. Inserting the expression (66) into his DP equation (64), E obtains the boundary conditions

$$\begin{aligned} P_N^{(E)} &= H, \\ p_N^{(E)} &= 0. \end{aligned}$$

In addition,

$$\begin{aligned} x_k^T P_k^{(E)} x_k + p_k^{(E)} &= \max_{v_k} E_{w_k} ([u_k^T R_u u_k - v_k^T R_v v_k + (Ax_k + Bu_k + Cv_k + \Gamma w_k)^T \\ &\quad \times (Q + P_{k+1}^{(E)})(Ax_k + Bu_k + Cv_k + \Gamma w_k) + p_{k+1}^{(E)}]) \\ &= \max_{v_k} [u_k^T R_u u_k - v_k^T R_v v_k + (Ax_k + Bu_k + Cv_k)^T (Q + P_{k+1}^{(E)}) \\ &\quad \times (Ax_k + Bu_k + Cv_k)] + \text{Trace}(\Gamma^T (Q + P_{k+1}^{(E)})\Gamma W) + p_{k+1}^{(E)}, \\ &\quad \forall k = 0, \dots, N-1. \end{aligned} \quad (70)$$

Upon performing the maximization operation and provided that

$$R_v - C^T(Q + P_k^{(E)})C > 0, \quad \forall k = 1, \dots, N, \quad (71)$$

E concludes that his optimal control v_k^* must satisfy the linear equation

$$[R_v - C^T(Q + P_{k+1}^{(E)})C]v_k^* - C^T(Q + P_{k+1}^{(E)})Bu_k = C^T(Q + P_{k+1}^{(E)})Ax_k. \quad (72)$$

Reassume P's point of view: P gets into the decision loop of player E and he figures that he must solve the set of linear equations (69) and (72) in u_k^* and v_k^* . P has all the information needed to solve the said system of equations.

Thus, similar to the analysis in Sect. 3, the optimal controls calculated by P are

$$\begin{pmatrix} u_k^* \\ v_k^* \end{pmatrix} = -(M_{k+1}^{(P,E)})^{-1} \begin{bmatrix} B^T(Q + P_{k+1}^{(P)})(Ax_k + \Gamma w_k) \\ C^T(Q + P_{k+1}^{(E)})Ax_k \end{bmatrix}, \quad (73)$$

with the $(m_u + m_v) \times (m_u + m_v)$ blocked matrix

$$M_{k+1}^{(P,E)} \equiv \begin{bmatrix} B^T(Q + P_{k+1}^{(P)})B + R_u & B^T(Q + P_{k+1}^{(P)})C \\ C^T(Q + P_{k+1}^{(E)})B & C^T(Q + P_{k+1}^{(E)})C - R_v \end{bmatrix}. \quad (74)$$

The matrix $M_{k+1}^{(P,E)}$ must be invertible. This is indeed the case, provided that the Schur complement of the block $C^T(Q + P_{k+1}^{(E)})C - R_v$ in the $M_{k+1}^{(P,E)}$ matrix, namely, the $r_u \times r_u$ matrix

$$\begin{aligned} S_B^{(P,E)}(P_{k+1}^{(P)}, P_{k+1}^{(E)}) \\ \equiv B^T(Q + P_{k+1}^{(P)})B + R_u \\ + B^T(Q + P_{k+1}^{(P)})C[R_v - C^T(Q + P_{k+1}^{(E)})C]^{-1}C^T(Q + P_{k+1}^{(E)})B, \end{aligned} \quad (75)$$

is invertible.

Guess. At this point, we are motivated to introduce the additional guess

$$P_k^{(P)} = P_k^{(E)} = P_k, \quad (76)$$

where the real symmetric matrices P_k are the solution of the difference equation (5) obtained in the deterministic case.

The $n \times n$ matrices P_k are the same as in the deterministic LQDG addressed in Sects. 2 and 3 and the stochastic LQGDG1 and LQGDG2, addressed in the respective Sects. 6 and 7, as will become apparent in the sequel.

This guess yields the requirement that the matrix $S_B(P_{k+1})$ [see e.g. (23)] be invertible.

Furthermore, the players' value functions will be

$$V_k^{(P)}(x_k) = x_k^T P_k x_k + p_k^{(P)}, \quad (77)$$

$$V_k^{(E)}(x_k) = x_k^T P_k x_k + p_k^{(E)}. \quad (78)$$

In addition, conditions (68) and (71) are equivalent to the respective conditions (3) and (4). Thus, everything falls into place and the matrix S_B is invertible.

Finally, (69) and (72) are now

$$[R_u + B^T(Q + P_{k+1})B]u_k^* + B^T(Q + P_{k+1})Cv_k = -B^T(Q + P_{k+1})(Ax_k + \Gamma w_k) \quad (79)$$

$$[C^T(Q + P_{k+1})C - R_v]v_k^* + C^T(Q + P_{k+1})Bu_k = -C^T(Q + P_{k+1})Ax_k, \quad (80)$$

so that

$$\begin{pmatrix} u_k^* \\ v_k^* \end{pmatrix} = -M_{k+1}^{-1} \begin{bmatrix} B^T(Q + P_{k+1})(Ax_k + \Gamma w_k) \\ C^T(Q + P_{k+1})Ax_k \end{bmatrix}, \quad (81)$$

and similar to the calculation in Sect. 3.2,

$$\begin{aligned} u_k^* &= -S_B^{-1}(P_{k+1})B^T\{I + (Q + P_{k+1})C[R_v - C^T(Q + P_{k+1})C]^{-1}C^T\}(Q + P_{k+1}) \\ &\quad \times A \cdot x_k \\ &\quad - S_B^{-1}(P_{k+1})B^T(Q + P_{k+1})\Gamma \cdot w_k. \end{aligned}$$

Using the forward recursion (5) for P_k and the definition $P_k := P_k + Q$, the linear state feedback-stroboscopic strategy of player P is obtained,

$$\begin{aligned} u_k^*(x_k, w_k) &= -S_B^{-1}(P_{N-k-1})B^T\{[I + P_{N-k-1}C(R_v - C^T P_{N-k-1}C)^{-1}C^T] \\ &\quad \times P_{N-k-1}A \cdot x_k + P_{N-k-1}\Gamma \cdot w_k\}. \end{aligned} \quad (82)$$

The strategy (82) is intuitively appealing. Compare with the P player strategy in LQGDG1, where neither player had access to the random input, (9) in Sect. 6, and the P player's strategy in LQGDG2, (58) in Sect. 7, where both players were privy to the Nature input.

Equation (81) also yields the strategy v^* of the E player. Similar to u^* , also v^* is a function of the random input w_k : the E player's strategy v^* is not realizable because the E player does not have access to Nature's input w_k ; this strategy is discarded.

The E player can however derive the P player optimal strategy (82). Hence, the E player can compute the optimal response to the P player strategy (82). To this end, E revisits (70) and calculates

$$\begin{aligned}
 x_k^T P_k x_k + p_k^{(E)} &= \max_{v_k} E_{w_k} ([u_k^T R_u u_k - v_k^T R_v v_k + (Ax_k + Bu_k + Cv_k + \Gamma w_k)^T \\
 &\quad \times (Q + P_{k+1})(Ax_k + Bu_k + Cv_k + \Gamma w_k) + p_{k+1}^{(E)}]) \\
 &= \max_{v_k} E_{w_k} ([u_k^T R_u u_k - v_k^T R_v v_k \\
 &\quad + (Ax_k + Bu_k + Cv_k)^T (Q + P_{k+1})(Ax_k + Bu_k + Cv_k)] \\
 &\quad + \text{Trace}(\Gamma^T (Q + P_{k+1}) \Gamma W) + p_{k+1}^{(E)}) \\
 &= \max_{v_k} \{[(Ax_k + Cv_k)^T (Q + P_{k+1})(Ax_k + Cv_k) - v_k^T R_v v_k] \\
 &\quad + 2E_{w_k}(u_k^T) B^T (Q + P_{k+1})(Ax_k + Cv_k)\} \\
 &\quad + \text{Trace}(\Gamma^T (Q + P_{k+1}) \Gamma W) + p_{k+1}^{(E)} \\
 &\quad + E_{w_k}(u_k^T [R_u + B^T (Q + P_{k+1}) B] u_k), \quad \forall k = 0, \dots, N-1,
 \end{aligned} \tag{83}$$

where u_k is given by (82). E calculates

$$\begin{aligned}
 &E_{w_k}(u_k^T) B^T (Q + P_{k+1})(Ax_k + Cv_k) \\
 &= -(Ax_k + Cv_k)^T (Q + P_{k+1}) B S_B^{-1} (P_{k+1}) B^T \\
 &\quad \times [I + (Q + P_{k+1}) C (R_v - C^T (Q + P_{k+1}) C)^{-1} C^T] (Q + P_{k+1}) Ax_k,
 \end{aligned}$$

so that the optimization must be performed,

$$\begin{aligned}
 &\max_{v_k} \{ (Ax_k + Cv_k)^T (Q + P_{k+1})(Ax_k + Cv_k) - v_k^T R_v v_k \\
 &\quad - 2(Ax_k + Cv_k)^T (Q + P_{k+1}) B S_B^{-1} (P_{k+1}) B^T \\
 &\quad \times [I + (Q + P_{k+1}) C (R_v - C^T (Q + P_{k+1}) C)^{-1} C^T] (Q + P_{k+1}) Ax_k \},
 \end{aligned}$$

that is

$$\begin{aligned}
 &\max_{v_k} \{ -v_k^T [R_v - C^T (Q + P_{k+1}) C] v_k + 2v_k^T C^T (Q + P_{k+1}) [I - B S_B^{-1} (P_{k+1}) B^T \\
 &\quad \times [I + (Q + P_{k+1}) C (R_v - C^T (Q + P_{k+1}) C)^{-1} C^T] (Q + P_{k+1})] Ax_k \}.
 \end{aligned}$$

Thus,

$$\begin{aligned}
 v_k^*(x_k) &= [R_v - C^T (Q + P_{k+1}) C]^{-1} C^T (Q + P_{k+1}) \{ I - B S_B^{-1} (P_{k+1}) B^T \\
 &\quad \times [I + (Q + P_{k+1}) C (R_v - C^T (Q + P_{k+1}) C)^{-1} C^T] (Q + P_{k+1}) \} A \cdot x_k.
 \end{aligned}$$

Using the forward recursion (5) for P_k and the definition $P_k := P_k + Q$, the linear state feedback strategy of player E is obtained,

$$\begin{aligned}
 v_k^*(x_k) &= [R_v - C^T P_{N-k-1} C]^{-1} C^T P_{N-k-1} \{ I - B S_B^{-1} (P_{N-k-1}) B^T \\
 &\quad \times [I + P_{N-k-1} C (R_v - C^T P_{N-k-1} C)^{-1} C^T] P_{N-k-1} \} A \cdot x_k. \tag{84}
 \end{aligned}$$

One thus recovers E's optimal strategy from Sects. 2 and 6, where both players had access to the system state, but not to Nature's input [see, e.g., (54)]. Furthermore, inserting the optimal strategies (82) and (84) into player E's DP equation (83), we obtain the following recursion for the scalar sequence

$$\begin{aligned} p_{k+1}^{(E)} &= p_k^{(E)} + \text{Trace}(\Gamma^T [P_k + P_k B S_B^{-1}(P_k)(R_u + B^T P_k B) S_B^{-1}(P_k) B^T P_k] \Gamma W), \\ p_0^{(E)} &= 0, \quad k = 0, \dots, N-1. \end{aligned} \quad (85)$$

Similarly, starting out from the DP recursion (67), it can be shown that the optimal response of player P to E's state feedback strategy (84) ($\equiv$ (54)) is the state feedback-stroboscopic strategy (82). Furthermore, inserting the optimal strategies (82) and (84) into player P's DP equation (67), we obtain the following recursion for the scalar sequence

$$\begin{aligned} p_{k+1}^{(P)} &= p_k^{(P)} + \text{Trace}(\Gamma^T [P_k + P_k B S_B^{-1}(P_k)(R_u + B^T P_k B) S_B^{-1}(P_k) B^T P_k] \Gamma W), \\ p_0^{(P)} &= 0, \quad k = 0, \dots, N-1. \end{aligned} \quad (86)$$

Equations (85) and (86) are identical, wherefrom we conclude that

$$p_k^{(P)} = p_k^{(P)} \equiv p_k^{(3)}, \quad \forall k = 0, \dots, N-1.$$

Hence, it is remarkable that also the LQGDG3 is a zero-sum game.

These results are summarized in the following theorem.

Theorem 8.1 *Consider the stochastic control system (47) and payoff functional (48). Assume that at decision time k the player P has access to the state x_k and the random disturbance w_k whereas player E has access to the state x_k only. In the LQGDG3, P's optimal strategy is given by the state feedback/stroboscopic control law (82) and E's optimal strategy is given by the state feedback control law (54). The LQGDG3 is a zero-sum game and its value function is*

$$V_0^{(3)}(x_0) = x_0^T (P_N - Q) x_0 + p_N^{(3)}. \quad (87)$$

The real symmetric matrices P_k are calculated according to the recursion (5) for the deterministic LQDG, and having obtained the sequence $\{P_k\}_{k=0}^N$, the new scalar sequence is calculated according to the recursion

$$\begin{aligned} p_{k+1}^{(3)} &= p_k^{(3)} + \text{Trace}(\Gamma^T [P_k + P_k B S_B^{-1}(P_k)(R_u + B^T P_k B) S_B^{-1}(P_k) B^T P_k] \Gamma W), \\ p_0^{(3)} &= 0, \quad k = 0, \dots, N-1. \end{aligned} \quad (88)$$

As in the deterministic case, conditions (3)–(4) are necessary and sufficient for the existence of a solution to the zero-sum LQGDG3 on the planning horizon N .

Similar results can be obtained when the roles are reversed and player E has the information advantage. New scalar sequences $p^{(1)}$, $p^{(2)}$ and $p^{(3)}$ must be calculated.

9 Conclusions

In this article a hierarchy of linear-quadratic dynamic games is discussed, in the context of the application of the theory of dynamic games to optimal disturbance rejection using digital signal processing. Specifically, *discrete-time*, finite-dimensional, LQDGs are considered. In discrete time the cost function contains product terms between the decision variables, which complicates the analysis compared to its continuous time analogue. With a view to facilitating the application of the theory of dynamic games to digital signal processing and in particular disturbance rejection, the complete solution of discrete-time LQDGs is worked out, and explicit results are derived. Special attention is given to the infinite-horizon discrete-time LQDG in order to obtain linear time invariant results for steady-state disturbance rejection action. The results are illustrated in the example discussed in Sect. 4, where some fine points concerning dynamic games are highlighted. The presentation is tutorial in nature, yet the application of the theory of dynamic games to active disturbance rejection in control systems has the potential to significantly enrich both fields. The disturbance rejection paradigm serves as a vehicle for investigating the impact of the information pattern on the solution of dynamic games. Building on our explicit solution of deterministic LQDGs, a hierarchy of three zero-sum stochastic LQDGs characterized by information patterns which increase in complexity is analyzed, yielding insight into the role that information plays in dynamic games.

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